



MAARULA CLASSES

FOR DETAILED SOLUTION VISIT

www.maarula.in

MAARULA MATHEM E-LEARNING APP

Solved With Tricks

MATRICES **NIMCET PYQ**

Scan The QR Download
Our App Now.



➡ **Topic Wise Weightage &**

Detailed Trend Analysis

➡ **All 17 years Paper**

With Smart Solution

➡ **Syllabus & Important**

Documents

➡ **Topic Wise All PYQ'S**

NIMCET-Free Test



By Amit Katiyar (MCA-JNU)



1. If $A = \begin{bmatrix} 1 & 2 & 15 \\ 3 & 4 & 11 \\ 5 & 6 & 7 \end{bmatrix}$, let $\lambda_1, \lambda_2, \lambda_3$ be its eigen value then
 $(1 + \lambda_1)(1 + \lambda_2)(1 + \lambda_3) =$ **NIMCET 2007**
 (a) 0 (b) -95 (c) -4 (d) 12

2. $A = \begin{pmatrix} 0 & 0 & -1 \\ 0 & -1 & 0 \\ -1 & 0 & 0 \end{pmatrix}$ The only correct statement about the matrix A is: **NIMCET 2007**
 (a) A^{-1} does not exist
 (b) $A = (-1)I$, where I is a unit matrix
 (c) A is a zero matrix
 (d) $A^2 = I$

3. If $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$, then $I + A + A^2 + \dots \infty$ equal to **NIMCET 2009**

- (a) $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ (b) $\begin{bmatrix} -1 & -2 \\ -3 & -4 \end{bmatrix}$
 (c) $\begin{bmatrix} \frac{1}{2} & -\frac{1}{3} \\ -\frac{1}{2} & 0 \end{bmatrix}$ (d) $\begin{bmatrix} -\frac{1}{4} & \frac{1}{3} \\ \frac{1}{2} & 0 \end{bmatrix}$

4. If A is a 3×3 matrix with $\det(A) = 3$, then $\det(\text{adj } A)$ is **NIMCET 2009**
 (a) 3 (b) 9
 (c) 27 (d) 6

5. If A and B are two square matrices such that $B = -A^{-1}BA$, then $(A + B)^2$ will be equal to **NIMCET 2011**
 (a) 0 (b) $A^2 + 2A + B^2$
 (c) $A^2 + B^2$ (d) $A + B$

6. If $A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$, then A^n for any natural number n is **NIMCET 2012**

- (a) $\begin{bmatrix} n & n \\ 0 & n \end{bmatrix}$ (b) $\begin{bmatrix} 1 & n \\ 0 & 1 \end{bmatrix}$
 (c) $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ (d) None of these

7. If $A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & -2 & -4 \end{bmatrix}$ and $6A^{-1} = A^2 + cA + dI$, where A^{-1} is the inverse of A, I is the identity matrix, then (c, d) is **NIMCET 2013**

- (a) (-6, 11) (b) (6, -11)
 (c) (11, -6) (d) (6, 11)

8. If the matrix $\begin{bmatrix} -1 & 3 & 2 \\ 1 & K & -3 \\ 1 & 4 & 5 \end{bmatrix}$ has an inverse matrix, then the value of K is **NIMCET 2014**
 (a) K is any real number (b) $K \neq -4$
 (c) $k = -4$ (d) $k \neq 4$

9. If $A = \begin{bmatrix} a & b & c \\ b & c & a \\ c & a & b \end{bmatrix}$, where a, b and c are real positive numbers such that $abc = 1$ and $A^T A = I$, then the equation that holds true among the following is **NIMCET 2015**

- (a) $a + b + c = 1$ (b) $a^2 + b^2 + c^2 = 1$
 (c) $ab + bc + ca = 0$ (d) $a^3 + b^3 + c^3 = 4$

10. The matrix A has x rows and $(x + 5)$ columns and the matrix B has y rows and $(11 - y)$ columns. If both the matrices AB and BA exist, then the value of x and y are **NIMCET 2016**

- (a) 8, 3 (b) 3, 5 (c) 3, 8 (d) 8, 5

11. Find a matrix X such that $2A + B + X = 0$, where $A = \begin{bmatrix} -1 & 2 \\ 3 & 4 \end{bmatrix}$ and $B = \begin{bmatrix} 3 & -2 \\ 1 & 5 \end{bmatrix}$ **NIMCET 2017**

- (a) $\begin{bmatrix} 1 & 2 \\ 7 & 13 \end{bmatrix}$ (b) $\begin{bmatrix} -1 & -2 \\ -7 & -13 \end{bmatrix}$
 (c) $\begin{bmatrix} 13 & 2 \\ 7 & 1 \end{bmatrix}$ (d) $\begin{bmatrix} -13 & -2 \\ -7 & -1 \end{bmatrix}$

12. If $A = \begin{bmatrix} 0 & 5 \\ 0 & 0 \end{bmatrix}$ and $f(x) = 1 + x + x^2 + \dots x^{16}$, then $f(A)$ is equal to **NIMCET 2018**
 (a) 0 (b) $\begin{bmatrix} 1 & 5 \\ 0 & 1 \end{bmatrix}$ (c) $\begin{bmatrix} 1 & 5 \\ 0 & 0 \end{bmatrix}$ (d) $\begin{bmatrix} 0 & 5 \\ 1 & 1 \end{bmatrix}$

13. If A is an invertible skew symmetric matrix, then A^{-1} is a **NIMCET 2018**
 (a) symmetric matrix (b) skew-symmetric matrix
 (c) zero matrix (d) identify matrix

14. If $A = \begin{bmatrix} \cos \alpha & \sin \alpha \\ -\sin \alpha & \cos \alpha \end{bmatrix}$, then for any positive integer n, A^n is **NIMCET 2020**

- (a) $\begin{bmatrix} \sin n\alpha & \cos n\alpha \\ \cos n\alpha & -\sin n\alpha \end{bmatrix}$ (b) $\begin{bmatrix} \cos n\alpha & \sin n\alpha \\ \sin n\alpha & \cos n\alpha \end{bmatrix}$
 (c) $\begin{bmatrix} \cos n\alpha & \sin n\alpha \\ \sin n\alpha & -\cos n\alpha \end{bmatrix}$ (d) $\begin{bmatrix} \cos n\alpha & \sin n\alpha \\ -\sin n\alpha & \cos n\alpha \end{bmatrix}$

15. Let $A = (a_{ij})$ and $B = (b_{ij})$ be two square matrices of order n and $\det(A)$ denotes the determinant of A. then, which of the following is not correct. **NIMCET 2020**

- (a) If A is diagonal matrix, then $\det(A) = a_{11}a_{22} \dots a_{nn}$
 (b) $\det(AB) = \det(A) \det(B)$
 (c) $\det(CA) = C \det(A)$
 (d) $\det(A) = \det(A^T)$, where A^T denotes the transpose pf the matrix A



16. If $F(\theta) = \begin{bmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$, then find $F(\theta)F(\alpha)$ is

equal to

NIMCET 2021

- (a) $f(\theta\alpha)$ (b) $f(\theta/\alpha)$
(c) $f(\theta + \alpha)$ (d) $f(\theta - \alpha)$

17. If A and B are square matrices such that $B = -A^{-1}BA$, then $(A + B)^2$ is

NIMCET 2023

- (a) 0 (b) $A^2 + B^2$
(c) $A^2 + 2AB + B^2$ (d) $A + B$

18. The system of equations $x + 2y + 2z = 5$, $x + 2y + 3z = 6$, $x + 2y + \lambda z = \mu$ has infinitely many solution if (Matrix)

- (a) $\lambda \neq 2$ (b) $\lambda \neq 2, \mu \neq 5$
(c) $\lambda = 2, \mu = 5$ (d) $\mu \neq 5$

NIMCET 2024

19. Let A and B be two square matrices of same order satisfying $A^2 + 5A + 5I = 0$ and $B^2 + 3B + I = 0$ respectively. Where I is the identity matrix. Then the inverse of the matrix $C = BA + 2B + 2A + 4I$

NIMCET 2025

- (a) $AB + A + 3B + 3I$ (b) $BA + 3B + A + 3I$
(c) $BA - 3B + A - 3I$ (d) $AB - A + 3B - 3I$

20. Consider the matrix $B = \begin{pmatrix} -1 & -1 & 2 \\ 0 & -1 & -1 \\ 0 & 0 & -1 \end{pmatrix}$. The sum of all the entries of the matrix B^{19} is

NIMCET 2025

- (a) -171 (b) -192
(c) -174 (d) -163

ANSWER KEY

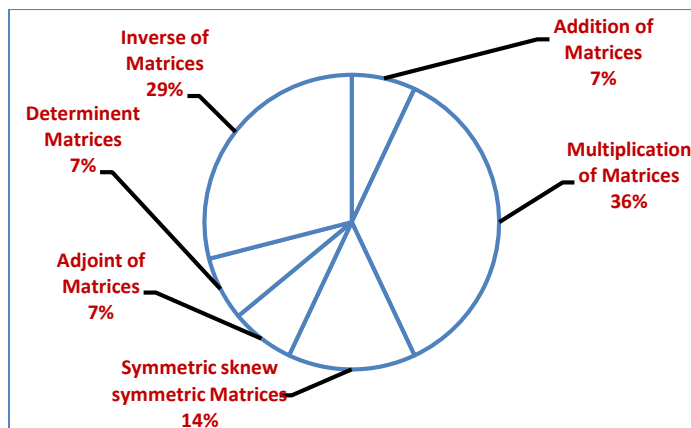
1.	2.	3.	4.	5.	6.	7.	8.	9.	10.
b	d	c	b	c	b	a	b	b	c
11.	12.	13.	14.	15.	16.	17.	18.	19.	20.
b	b	b	d	c	c	b	c	a	c

Analysis 2007 to 2025

Questions Asked in NIMCET 2007 to 2025

Years	No. of Questions	Years	No. of Questions
NIMCET 2007	02	NIMCET 2016	01
NIMCET 2008	00	NIMCET 2017	01
NIMCET 2009	02	NIMCET 2018	02
NIMCET 2010	00	NIMCET 2019	00
NIMCET 2011	01	NIMCET 2020	02
NIMCET 2012	01	NIMCET 2021	01
NIMCET 2013	01	NIMCET 2022	00
NIMCET 2014	01	NIMCET 2023	01
NIMCET 2015	01	NIMCET 2024	01
		NIMCET 2025	02

Sub Topic Analysis



Sub Topicwise Analysis

Sub topic	Question Asked
Addition of Matrices	01
Multiplication of Matrices	05
Symmetric Skew Symmetric Matrices	02
Determinant of Matrices	01
Adjoint of Matrices	01
Inverse of Matrices	05

SOLUTION

1. (b)

$$A = \begin{bmatrix} 1 & 2 & 15 \\ 3 & 4 & 11 \\ 5 & 6 & 7 \end{bmatrix}, \text{ let } \lambda_1, \lambda_2, \lambda_3 \text{ be eigen value}$$

For eigen value

$$|A - \lambda I| = 0$$

$$\begin{vmatrix} 1-\lambda & 2 & 15 \\ 3 & 4-\lambda & 11 \\ 5 & 6 & 7-\lambda \end{vmatrix} = 0$$

$$(1-\lambda)[(4-\lambda)(7-\lambda) - 66] - 2(21 - 3\lambda - 55) + 15(8 - 20 + 5\lambda)$$

$$(1-\lambda)(4-\lambda)(7-\lambda) - 66 + 66\lambda + 2 \times 34 + 6\lambda + 75\lambda + 30$$

$$(1-\lambda)(4-\lambda)(7-\lambda) - 66 + 68 + 130 + 72\lambda + 75\lambda$$

$$28 - \lambda^3 + 12\lambda^2 - 39\lambda + 28 + 147\lambda = 0$$

$$\lambda^3 - 12\lambda^2 - 39\lambda - 108\lambda = 0$$

$$\lambda = -95.$$

2. (d) $A = \begin{bmatrix} 0 & 0 & -1 \\ 0 & -1 & 0 \\ -1 & 0 & 0 \end{bmatrix}$

(i) $|A| = 1$ A^{-1} exist

(ii) $A \neq (-1)I$

(iii) A is not a zero matrix

(iv) $A^2 = I$

$$A^2 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = I$$

3. (c) $I + A + A^2 + \dots \dots \infty$

$$= \frac{a}{1-r} = \frac{I}{1-A} = I(1-A)^{-1} = (1-A)^{-1}$$



$$1 - A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} - \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} = \begin{bmatrix} 0 & -2 \\ -3 & -3 \end{bmatrix}$$

$$(1 - A)^{-1} = \begin{bmatrix} 0 & -2 \\ -3 & -3 \end{bmatrix}^{-1} = T$$

$$\text{Adj}T = \begin{bmatrix} -3 & 2 \\ 3 & 0 \end{bmatrix}$$

$$T^{-1} = \frac{\text{adj}T}{|T|} = \frac{-1}{6} \begin{bmatrix} -3 & 2 \\ 3 & 0 \end{bmatrix} = \begin{bmatrix} \frac{1}{2} & -\frac{1}{3} \\ -\frac{1}{2} & 0 \end{bmatrix}$$

4. (b) $|\text{adj} A| = |A|^{n-1} = 3^{n-1} = 3^2 = 9.$

5. (c)

$$-A^{-1}BA$$

Multiplying both sides by matrix A

$$AB = -AA^{-1}BA$$

$$\text{As we know } AA^{-1} = I$$

$$AB = -BA$$

$$(A + B)^2 = A^2 + AB + BA + B^2$$

$$= A^2 + AB - AB + B^2 = A^2 + B^2$$

6. (b) $A^2 = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}$

$$A^3 = A^2 \cdot A = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 3 \\ 0 & 1 \end{bmatrix}$$

$$\text{Similarly, } A^n = \begin{bmatrix} 1 & n \\ 0 & 1 \end{bmatrix}$$

7. (a) From characteristic equation $|A - \lambda I| = 0$

$$\begin{vmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & -2 & 4 \end{vmatrix} - \begin{vmatrix} \lambda & 0 & 0 \\ 0 & \lambda & 0 \\ 0 & 0 & \lambda \end{vmatrix} = 0$$

$$\Rightarrow \begin{vmatrix} 1-\lambda & 0 & 0 \\ 0 & 1-\lambda & 1 \\ 0 & -2 & 4-\lambda \end{vmatrix} = 0$$

$$\Rightarrow \lambda^3 - 6\lambda^2 + 11\lambda - 6 = 0$$

This is our characteristics equation

$$\Rightarrow A^3 - 6A^2 + 11A - 6 = 0$$

Multiply by A^{-1}

$$\Rightarrow (A^{-1}A)A^2 - 6(A^{-1}A)A + 11A^{-1}A - 6A^{-1} = 0$$

$$\Rightarrow A^2 - 6A + 11I = 6A^{-1}$$

$$\Rightarrow 6A^{-1} = A^2 - 6A + 11I$$

$$\text{Hence } C = -6, d = 11$$

8. (b)

If a inverse matrix A exist, then $|A| \neq 0$

$$\begin{vmatrix} -1 & 3 & 2 \\ 1 & K & -3 \\ 1 & 4 & 5 \end{vmatrix} \neq 0$$

$$-1(5K + 12) - 3(5 + 3) + 2(4 - k) \neq 0$$

$$-5K - 12 - 24 - 2K + 8 \neq 0$$

$$K \neq -4$$

9. (b) Because of A is a symmetric matrix.

$$\text{So, } A^T = A$$

$$A \cdot A^T = I$$

$$\begin{bmatrix} a & b & c \\ b & c & a \\ c & a & b \end{bmatrix} \begin{bmatrix} a & b & c \\ b & c & a \\ c & a & b \end{bmatrix} = I$$

$$\begin{bmatrix} a^2 + b^2 + c^2 & ab + bc + ca & ac + ab + bc \\ ab + bc + ac & b^2 + c^2 + a^2 & bc + ca + ab \\ ac + ab + bc & bc + ca + ab & c^2 + a^2 + b^2 \end{bmatrix} =$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$a^2 + b^2 + c^2 = 1$$

$$ab + bc + ca = 0 [\because \text{Not possible } a, b, c \text{ real and postive}]$$

10. (c)

If AB possible, then number. of column in A

= Number of rows in B

$$x + 5 = y \Rightarrow x - y = -5$$

If BA possible, the then number of column in B

= Number of rows in A

$$11 - y = x$$

$$\Rightarrow x + y = 11$$

Solve both Eqs. (i) and (ii)

$$x - y = -5$$

$$2x = 6$$

$$3 - y = -5$$

$$x + y = 11$$

$$x = 3 \quad y = 8$$

11. (b) $2A + B = \begin{bmatrix} -2 & 4 \\ 6 & 8 \end{bmatrix} + \begin{bmatrix} 3 & -2 \\ 1 & 5 \end{bmatrix} = \begin{bmatrix} 1 & 2 \\ 7 & 13 \end{bmatrix}$

$$\Rightarrow X = -(2A + B) \Rightarrow X = \begin{bmatrix} -1 & -2 \\ -7 & -13 \end{bmatrix}$$

12. (b) $A^2 = \begin{bmatrix} 0 & 5 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 0 & 5 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$

So, A^2 and all higher powers of matrix A will be null matrix.

$$f(A) = 1 + A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} + \begin{bmatrix} 0 & 5 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 5 \\ 0 & 1 \end{bmatrix}$$

13. (b) If A is a skew symmetric matrix of odd order, then inverse does not exist.

$\Rightarrow$ If A is a skew symmetric matrix of even order, then A^{-1} is also a skew symmetric matrix.

14. (d) $A^2 = \begin{bmatrix} \cos \alpha & \sin \alpha \\ -\sin \alpha & \cos \alpha \end{bmatrix} \begin{bmatrix} \cos \alpha & \sin \alpha \\ -\sin \alpha & \cos \alpha \end{bmatrix}$

$$= \begin{bmatrix} \cos^2 \alpha - \sin^2 \alpha & 2\sin \alpha \cos \alpha \\ -2\sin \alpha \cos \alpha & \cos^2 \alpha - \sin^2 \alpha \end{bmatrix}$$

$$= \begin{bmatrix} \cos 2\alpha & \sin 2\alpha \\ -\sin 2\alpha & \cos 2\alpha \end{bmatrix}$$

$$A^2 \cdot A = A^3 = \begin{bmatrix} \cos 2\alpha & \sin 2\alpha \\ -\sin 2\alpha & \cos 2\alpha \end{bmatrix} \begin{bmatrix} \cos \alpha & \sin \alpha \\ -\sin \alpha & \cos \alpha \end{bmatrix}$$



$$= \begin{bmatrix} \cos 2\alpha \cdot \cos \alpha - \sin 2\alpha \sin \alpha & \cos 2\alpha \sin \alpha + \sin 2\alpha \cos \alpha \\ -(\sin 2\alpha \cos \alpha + \cos 2\alpha \sin \alpha) & \cos 2\alpha \cos \alpha - \sin 2\alpha \sin \alpha \end{bmatrix}$$

$$= \begin{bmatrix} \cos(2\alpha + \alpha) & \sin(2\alpha + \alpha) \\ -\sin(2\alpha + \alpha) & \cos(2\alpha + \alpha) \end{bmatrix}$$

$$= \begin{bmatrix} \cos 3\alpha & \sin 3\alpha \\ -\sin 3\alpha & \cos 3\alpha \end{bmatrix}$$

Similarly, $A^n = \begin{bmatrix} \cos n\alpha & \sin n\alpha \\ -\sin n\alpha & \cos n\alpha \end{bmatrix}$

15. (c)

(a) It is true that if A is any diagonal matrix, then determinant of A is multiplication of diagonal elements.

(b) $\det(AB) = \det(A) \cdot \det(B)$

[By property of determinant]

(c) $\det(CA) = C\det(A)$; This is wrong because property is $|CA| = C^n|A|$.

(d) $\det(A) = \det(A^T)$ [By property of determinant]

16. (c) $f(\alpha) = \begin{bmatrix} \cos \alpha & -\sin \alpha & 0 \\ \sin \alpha & \cos \alpha & 0 \\ 0 & 0 & 1 \end{bmatrix}$

$$f(\theta) \cdot f(\alpha) = \begin{bmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \cos \alpha & -\sin \alpha & 0 \\ \sin \alpha & \cos \alpha & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} \cos \theta \cos \alpha - \sin \theta \sin \alpha & -(\cos \theta \sin \alpha + \sin \theta \cos \alpha) & 0 \\ \sin \theta \cos \alpha + \cos \theta \sin \alpha & \cos \theta \cos \alpha - \sin \theta \sin \alpha & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} \cos(\theta + \alpha) & -\sin(\theta + \alpha) & 0 \\ \sin(\theta + \alpha) & \cos(\theta + \alpha) & 0 \\ 0 & 0 & 1 \end{bmatrix} = f(\theta + \alpha)$$

17. (b)

$B = -A^{-1}BA$

$AB = -(AA^{-1})BA$

$AB = -BA$

$(A + B)^2 = A^2 + B^2 + AB + BA = A^2 + B^2$

18. (c) $x + 2y + 2z = 5$

$x + 2y + 3z = 6$

$x + 2y + \lambda z = \mu$

$Ax = B$

$$\begin{bmatrix} 1 & 2 & 2 & : & 5 \\ 1 & 2 & 3 & : & 6 \\ 1 & 2 & \lambda & : & \mu \end{bmatrix}$$

$R_2 \rightarrow R_2 - R_1$ and $R_3 \rightarrow R_3 - R_1$

$$\begin{bmatrix} 1 & 2 & 2 & : & 5 \\ 0 & 0 & 1 & : & 1 \\ 0 & 0 & \sqrt{-2} & : & \mu - 5 \end{bmatrix}$$

For many solution, Rank (AB) order of matrix

So $\lambda - 2 = 0$, $\lambda = 2$, $\mu - 5 = 0$, $\mu = 5$

19. (a)

Given $A^2 = -5A - 5I$ and $B^2 = -3B - I$.

Let $C = BA + 2B + 2A + 4I$, $X = AB + A + 3B + 3I$

We want $CX = I$. Expand CX and immediately replace any A^2 or B^2 using the given relations:

\$\$

$CX = (BA + 2B + 2A + 4I)(AB + A + 3B + 3I)$

$= BA \cdot AB + BA \cdot A + 3BA \cdot B + 3BA$

$+ 2B \cdot AB + 2B \cdot A + 6B^2 + 6B$

$+ 2A \cdot AB + 2A^2 + 6AB + 6A$

$+ 4AB + 4A + 12B + 12I$.

\$\$

Now substitute

$A^2 = -5A - 5I$ and $B^2 = -3B - I$ wherever they occur, and collect like terms. After substitution the non-scalar matrix terms cancel pairwise and the remaining scalar terms combine to give exactly I . Hence

\$\$

$CX = I, \$\$$

so X is the inverse of C . Therefore the inverse is

$AB + A + 3B + 3I$,

option (a).

20. (c)

Let

$B = \begin{pmatrix} -1 & -1 & 2 \\ 0 & -1 & -1 \\ 0 & 0 & -1 \end{pmatrix} = -I + N, N = \begin{pmatrix} 0 & -1 & 2 \\ 0 & 0 & -1 \\ 0 & 0 & 0 \end{pmatrix}, N^3 = 0.$

So,

$B^{19} = (-I + N)^{19} = -I + 19N - 171N^2,$

$N^2 = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$

Sum of entries is linear, hence

$\text{Sum } B^{19} = \text{Sum } (-I) + 19 \text{Sum}(N) - 171 \text{Sum}(N^2)$

$= -3 + 19 \cdot 0 - 171 \cdot 1$

$= -174.$

